



Forget the Password: Password Memory and Security Applications of Augmented Cognition

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Abstract. Individual security behavior plays a central role in achieving secure computing. However, secure usage is difficult to guarantee in an open-ended context where different users have different perceptions of security as well as different cognitive loads when using security tools. In designing secure systems, it is not only necessary to define secure behavior but also to provide built-in support for such behavior in order to enable users to be complaint. In this work, we explore the viability of augmented cognition as a modality that can be used to support security-oriented behavior in authentication systems. Specifically, we explore how transformations of password character properties such as font and weight can improve password recall and recognition and reduce insecure habits, such as writing down passwords. In a previous study, we tested the accuracy of recall and recognition in an augmented password system. The system was designed to make use of character property transformations to minimize the need for complex passwords while not compromising security. Here we repeat the study, incorporating the use of neurophysiological measures to study human physiological responses during recognition and recall of character sets with different types of transformation. The results suggest that cognitive effort in recall of complex passwords can be alleviated with the performance of the augmented password task. This finding has important implications for future research.

Keywords: Augmented cognition · Recall · Recognition · Password memory
Physiological measures · Authentication · Cybersecurity

1 Introduction

The pervasive presence of information technology has increased the information processing demands on the human cognitive system, elevating information intake, requiring faster assessment and more accurate decision-making [1]. On the other hand, technology is yet to sufficiently augment human cognition to meet the new expectations. Password authentication is a perfect case for this imbalanced scenario between cognitive processing expectations and human ability. Password authentication falls under the “what you know” paradigm of authentication which is commonly used due to lower cost and ease implementation compared to other methods. Password authentication is dependent on two cognitive processes namely recall and recognition [2]. Cabeza et al. [3] conducted an experiment to determine which parts of the brain are activated by recognition and recall processes. Their experiment involved the use of positron emission tomography (PET) to take measurements while participants recognized or recalled previously studied word pairs mixed with words not previously studied. They found that both processes caused observable activations in regions of the brain, with activation levels in the frontal cortex being indistinguishable between the two processes.

Recall and recognition are demanding on the human cognitive system and users typically have trouble remembering good passwords [4]. The result of this problem is that users typically choose weak passwords for easier memorization. For example, Florencio and Herley [5] conducted a study covering half a million accounts on the Internet over a period of 3 months. They found that when the participants were unconstrained, the majority created passwords that contained only lower case letters leaving out uppercase letters, digits, and special characters (modifications that make passwords more difficult to crack). The passwords they created represented an average bit strength of 40.54 bits, making them easy to crack in dictionary attacks. Additionally, the average password is reused in at least six different websites. In an attempt to improve password memory, different memory training techniques have been used including password rehearsal games [4].

Melby-Lervåg and Hulme [6], in their meta-analysis of memory training studies, concluded that memory training programs can yield reliable improvements on both verbal and nonverbal working memory tasks. However, these effects were likely to be short-term with minimal near-transfer effects. Near-transfer effects are those reflected in tasks similar to the ones in the training program [6]. Given the difficulties involved in improving cognitive ability, methods that aid the user’s cognitive processes during the use of a system might be more reliable for promoting secure password behavior.

In this paper, we explore the use of augmented cognition strategies for the purpose of aiding recognition and recall of passwords. Specifically, we study the use of character transformations to make passwords unique and memorable. We examine the extent to which character transformations make the passwords easier to recognize and recall. This strategy would also expand the password space within shorter passwords and hence reduce the need for longer and more complex ones that are difficult to use. In addition, we examine the neurophysiological correlates of password recognition and

recall using electroencephalography (EEG). We conclude with future avenues for augmented cognition work in password recognition and recall.

2 Related Work

2.1 Password Security Habits

Several studies have shown that typical computer users' password habits do not promote security [5, 8, 9]. Florencio and Herley [5] conducted a study covering half a million accounts online over a period of 3 months and found that majority of users when not restricted, chose passwords that contained only lower case letters leaving out uppercase letters, digits, and special characters. In another study conducted within a university setting, results showed that majority of the surveyed users used uppercase letter and numbers in their password formulation but typically used the numbers at the end of the string [9]. As the authors discuss, numbers placed at the end of the password string make it easier for attackers to build password cracking dictionaries, and 60% of the passwords they studied were breakable within days. Interestingly, this study found that there was a correlation between user self-rated security awareness and password strength. What they found was that users tend to overrate their security awareness [9]. The authors interpreted this as potentially resulting from misunderstanding of security concepts [9].

As Alohalı et al. [10] found in their study on human security behavior, several factors influence the likelihood that a user will practise good security habits. These factors include: cultural perception of security, technical savviness, awareness of security risk, security tool usability, past experiences with security incidents, individual cognition models and the user ability to understand risk, as well as personality traits and attitudes towards risk messages [10].

2.2 Information Overload and Cognition

Human actions can be understood in terms of dual process cognition with most of our actions being completed with little or no cognition (i.e., Kahneman's System 1 cognition) [12]. The study of cognitive sciences helps define what the human brain can do as well as where humans excel and the extent to which cognition can be influenced [11]. Password recall and recognition could be described in the context of problem solving which has been found to impose cognitive load on the memory [14].

Woods and Siponen [7] in their contextualized metamemory theory, suggest that the password recall problem may not be solely an indication of poor memory performance, but may also be caused by users' inaccurate perception of their own memory. In their study on password recall, memory performance, and metamemory, they found no significant correlation between memory performance and correct password recall. They also did not find a correlation between digit span and correct, immediate, or long-term recall. They argue that users who believe they have more memory capacity or that they have more control over remembering their passwords have a better password correct recall rate. These findings may mean a solution to password recall problems lies in

designing password systems that give more control to a user, and increase their motivation to memorize passwords and influence their expectations they will be able to remember passwords.

Either way, humans behavior eventually suffers from cognitive limitation as suggested by cognitive load theory [13]. The cognitive limitations of users, in turn, influences security behavior [15]. Specifically, awareness of cognitive limitations (or perhaps overestimation of them) influences security behavior [7]. In order to make passwords easier to remember, password users may select characters and words that are familiar to them such as their own names or birth dates. In cases where a password system enforces rules that lead to passwords that are not easily memorable, users may resort to writing them down on paper so as to be able to refer later. In addition, once a password is learned, the password owner may tend to reuse it on as many systems as possible. As it turns out, poor password choices are not by themselves an indication of security ignorance but often a result of the pursuit for simplicity and ease of use of computing systems [16]. People tend to create easier passwords and reuse them so they do not fall victim to their cognitive limitations.

Due to these realities, cognitive load is a critical factor to consider in information security design. For information system users, a high information security workload comes into direct conflict with system usability and results in less security-oriented behavior [17]. Albretchsen [17] found that while awareness campaigns on their own did not cause a change in behavior, users were more motivated towards security conscious behavior when using a system that actively engaged them in the security process. Dorneich et al. [1] defined a framework for mitigating cognitive bottlenecks in systems by employing augmented cognition design. They identified cognitive bottlenecks as those factors that made it difficult for the user to process information in a correct and timely manner. These bottlenecks include excess information, lower information processing ability than the computer they are interacting with and function misallocation where the user is left with a task they are not cognitively able to handle [1]. Augmented cognition models allow for redesign of the information processing interaction in a way that lessens cognitive load and allows the user's preparation to be better aligned to the cognitive task requirement.

Augmented cognition has been defined as “a form of human-systems interaction in which a tight coupling between the user and computer is achieved via physiological and neurophysiological sensing of a user's cognitive state. “This paradigm leverages knowledge of cognitive state to precisely adapt user-system interaction in real time [18].

2.3 Improving Password Habits Using Augmented Cognition

In the context of password creation, memory and usage, one of the main cognitive bottlenecks is function misallocation [1]. Restrictively directing users into creating passwords that combine characters in complex ways leads to misallocation of the information processing tasks between the computer and the user. The user is then left with an enormous task of recalling complex passwords in order to keep the system secure. The adjustable autonomy approach suggested in Dorneich et al. [1], indicates a situation where the user would have more leeway to define their own cognitive task.

As it relates to password use, this would mean the option to use a simpler password as needed. However, simpler passwords present a security risk as they can be easily cracked using rainbow tables and dictionaries [9]. A possible middleground is an augmented password system that utilizes character transformations to expand the password space and hence reduce the complexity requirement on the user. Such a system is discussed in Mogire et al. [2]. The design is aimed at making the password usage easier both by reducing the need for character complexity and by increasing memorability. Albrechtsen [17], in a qualitative study conducted to interpret participants' experiences of information security, found that while awareness campaigns on their own did not change behavior, users were more motivated towards security conscious behavior when using a system that actively engaged the user in the security design. The process of character transformation engages the user more actively in the creation of the password and could potentially increase their ability to recall it later.

2.4 Augmented Passwords: A Study

In 2017, a study was conducted to test the cognitive experience in an augmented password system [2]. Specifically, the study was designed to determine accuracy in recognition and recall of passwords using different font styles. 150 participants were recruited from a large-enrollment introductory computer science course at a research extensive university.

For the study, a six-character string password was used. The second character of the password string was modified for the different groups. There were five formats of the second character: the non-modified character, bold, italicize, underline, and strike-through modification. The password was displayed on the projector twice on the first day, then erased. Students were then asked to select the password from a group of similar password suggestions to assess recognition.

Two days later, students were asked to enter the password from memory to assess recall. After entering the password, a survey was administered to determine the methods used to recall the password.

Results showed that students had a recognition rate of 70% for no font style (plain text), 76% for bold text, 74% for italicize text, 75% for underline text, and 86% for strikethrough text. When asked to recall, students were accurate: 64% for no font style, 93% for bold text, 91% for italicize text, 94% for underline text, 76% for strikethrough text. Performance improved between recognition and recall for bolded, italicized, and underlined text by 17–19%. Conversely, performance decreased between recognition and recall: for no font style (–6%), strikethrough (–9%). Augmented password recall accuracy was higher than the non-augmented passwords.

Due to the higher recognition rate of augmented passwords, the authors believe that font style augmentation helped the participant to focus on the string and accurately replicate it with ease. Similarly, the researchers inferred that augmented characters were more memorable due to distinctiveness and hence had a higher recall rate [2]. Based on the greater recall rate, the authors believe that augmented passwords could be used in practice. An advantage of using the augmented characters is that it expands the password option space of traditional passwords.

However, further investigation is necessary, particularly to observe the neurophysiology of engaging with the augmented passwords. Neurophysiology would prove useful in defining the extent to which character transformations aid the cognition processes of recall and recognition compared to transformations where they are absent. The study below investigates this question by examining the neurophysiological correlates of password recognition and recall.

3 Methodology

3.1 Participant Demographic

This study drew participants from two senior computer science courses at a research intensive university. Participation was voluntary and extra credit was given as compensation for those choosing to participate. Each of the courses had alternative extra credit tasks for participants that did not wish to take part in a study. Nineteen people participated, there were 7 female and 12 male participants.

3.2 Study Task

To determine recognition accuracy, we used a Sakai course management system to create an experiment website that allowed participants to input a six-character string that included either no augmentation or one augmented character. The character string was six characters long (RNGKBV) and included bold, italicize, underline, strike-through, and no modification. The set included 25 variations that were delivered to participants in a random order with the exception of the first and last questions. These questions were the same for all participants to test for primacy and recency effects in recall. Another design flipped the order of the first and last question to reverse the test for primacy and recency. This created two separate groups for the test. Participants were randomly assigned to each group. At the end the participants were asked to enter the first and last strings in the set of 25.

3.3 Protocol

The participants arrived to the lab and informed consent was obtained. They then sat at the experiment desk. They logged into the university e-learning system and navigated to the study that was set up as a course with participants added to it.

After these initial steps, two standard-size disposable sensors were applied to the forearms to measure heart rate. Two standard-sized disposable sensors were attached to the sole of the participant's foot to measure skin conductance. Two small sensors were applied just above the participant's left eyebrow to measure frown muscle activity. An Emotiv EEG data collection headset was placed over the head to measure brainwave activity. The task began once the psychophysiological data collection sensors were applied.

3.4 Data Cleaning and Preparation

EEG data will be cleaned and analyzed using EEGLab. One limitation of EEG is that cortical bioelectrical activity is extremely small in magnitude when compared to muscle movements across the head. Therefore, participant movement introduces artifacts of high-frequency and magnitude into the EEG data. These will be removed using two methods: EEGLab probability calculations and visual inspection. The EEGLab artifact rejection algorithm uses deviations in microvolts greater than three standard deviations from the mean to reject specific trials. However, additional artifacts are also apparent to the trained eye, so visual inspection of trials is essential in artifact removal.

Electrodermal and facial EMG data were aggregated to mean values per second using BIOPAC's AcqKnowledge software. Change scores will be calculated by subtracting the physiological level at the onset of each target statement during the online discussion from each subsequent second across a 6-s window.

3.5 ICA Analysis of EEG Data

Initially, an EEGLab ICA performs a Principal Components Analysis (PCA). At each electrode site the program assesses which of the other electrode sites account for the most variance in the signal. Taking these weighted values it then relaxes the orthogonality constraint of PCA to isolate individual components of activation [19, 20]. Each ICA component then represents a pattern of activation over the entire brain, not solely the activity present at a specific electrode. The number of independent components (ICs) depends on the number of electrodes in the dataset, as the algorithm is working in an N -dimensional space (where N is the number of electrodes). Most participants in the current study are expected to generate 14 distinct ICs, since our recording device has 14 electrodes [20].

Finally, using the K -means component of EEGLab the independent components at the individual level will be grouped into clusters containing similar components using procedures recommended by Delorme and Makeig [20]. This procedure clusters similar ICs based upon their latency, frequency, amplitude, and scalp distribution. Relevant clusters will be identified and a time-frequency decomposition will be performed to examine changes in event-related desynchronization of the alpha rhythm.

4 Results

The results show increased alpha attenuation during the password augmentation task, which in turn led to better recall of the password. 11 participants correctly recalled the password asked at the end of the task, indicating the augmented password worked for most participants in invoking memory of the password. Significant alpha attenuation was observed during the recognition task (Fig. 1) with participants showing increased cognition while performing the augmentation task as compared to the recall task. These findings suggest that password recall can be aided by augmenting the password characters.

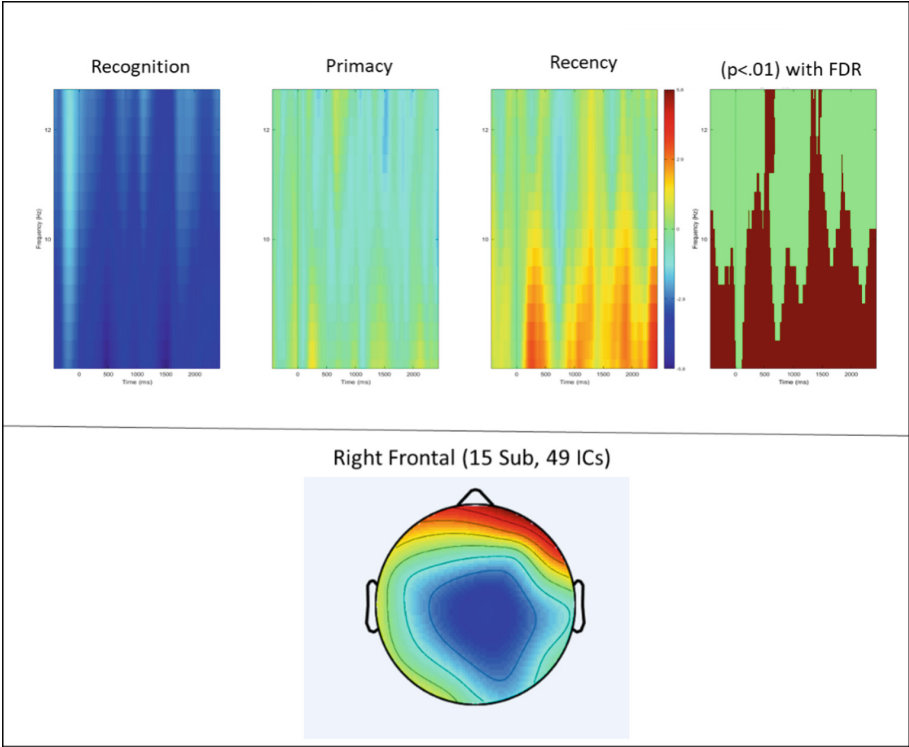


Fig. 1. Significant differences in the alpha band for recognition and recall (primacy and recency effects). Results indicate the password augmentation task increased cognition compared to the recall task.

We observed more alpha attenuation for the primacy than the recency recall condition across the participants measured (Fig. 1). This shows the participants required more cognition during recall of the first password augmentation than the last. Both recall conditions required less cognitive effort than the recognition task, indicating an alleviation of cognitive load during recall of a complex password. This finding indicates that this password augmentation task can aid in alleviating cognitive effort required for recall of complex passwords and could be useful in future applications.

5 Conclusion

Passwords still play an important role in security and poor user habits such as choosing easy-to-guess passwords, are driven by their desire to keep computer usage simple. Augmenting passwords strings using character font transformations can help improve users' cognition and memory of password. This can in turn motivate and promote security-conscious behavior in the password authentication process.

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